10 best examples and use cases of augmented reality in healthcare

While AI and ML are all the rage in healthcare, augmented reality is quietly becoming one of the industry's most practical, high-impact technologies. From operating rooms to physical therapy clinics, AR in healthcare opens up a completely new approach to medical education and even patient treatment. The numbers back it up: the global healthcare augmented and virtual reality market was valued at $3.40B in 2024, and now it's projected to skyrocket to $18.38 billion by 2034. Plus, AR dominates the market, as this technology alone accounts for 61% of it.

It's no wonder augmented reality for healthcare is consistently listed among top technology trends, and I believe we're only witnessing a tiny fraction of its true potential. In this article, I'll walk you through some of the best use cases of augmented reality in healthcare that make a tangible difference right now.

Key takeaways:

- AR is efficient at all stages of surgery, from pre-operative planning to executing and providing post-operative rehabilitation.
- The technology makes a difference in medical education and training, as well as in explaining to patients their conditions and upcoming procedures.
- Whether you want to implement AR, VR, XR, or 3D animations, Innowise is a reliable healthcare software development partner that can build even the most complex apps.

1. Immersive medical education and training

Instead of relying solely on textbooks or static 2D images, AR allows medical students and doctors in training to interact with detailed 3D models of human anatomy. It's one thing to read about the structure of the heart — it's another to walk around it, zoom in, rotate it, and understand how it works from all angles.

Sounds too good to be true? A study published in the Journal of Medical Education and Curricular Development proved that using medical animations improved students' test scores compared to regular books. Now, imagine the impact AR experience may have on the quality of knowledge in medical education. This promise really inspires my colleagues at Innowise to develop quality AR and VR solutions that will help shape practicing doctors of the future.

One of the strong examples of augmented reality in medicine I worked on is VOKA. It's a 3D atlas of normal anatomy and pathology, offering a hyper-realistic and interactive learning experience. What stands out to me is the VOKA AR mode: users can project 3D models into their real-world environment, making study sessions feel almost like hands-on dissections. You are welcome to check out the full article I wrote about this breakthrough project a few years back, or explore the video demo below to understand how the app works and what it brings.

VOKA 3D Anatomy & Pathology User Guide | Improve the way you study anatomy

2. Surgical planning and assistance

With AR tools, surgeons can now carefully study detailed, patient-specific anatomy models created using images like CT scans or MRIs. Not only does it help them plan the surgeries and navigate complex anatomy during minimally invasive operations, but improves clinical accuracy of each move.

While some members of the medical community may be sceptical about this technology, its efficiency has been proven. For instance, one study found that using mixed reality improves the accuracy of orbital reconstruction surgeries, as well as leads to an 8% increase in patient satisfaction. To me, this research shows that AR isn't here to replace a surgeon's skill but to sharpen it.

When speaking about AR surgery planning tools, OpenSight comes to mind. This platform is FDA-cleared and uses Microsoft HoloLens to project diagnostic imaging into the real world. OpenSight is designed for medical settings, enabling surgeons to view 2D, 3D, and even 4D images and plan surgeries accurately.

OpenSight Augmented Reality Surgical Planning | Novarad

3. Pre-operative visualization

Using AR in medical settings, surgeons overlay detailed 3D models of a patient's anatomy onto a patient's body. They get to see the actual pathology in a real-world environment, and this visual information helps them plan the timing, angles, and sequence of their actions before the surgery.

To my mind, the clear benefit of it, compared to scrolling through flat 2D images, is that AR is much more dynamic and visual. For sure, it doesn't magically solve surgical challenges, but it can definitely cut down time to make decisions and improve doctor's situational awareness before the surgery even begins.

One study examining the clinical efficiency of such a tool, Verima, caught my eye. Pediatric surgeons in the University Hospital in Bologna used the tool to project holograms of the pathologies on patients before the operations. Doctors participating in the study agreed that 3D visualization substantially improved their ability to interpret pathology dimensions and their relationships with the surrounding organs and vessels.

4. Intraoperative navigation

In an operating room, precision is everything, and that's why intraoperative AR navigation feels to me like one of the most natural applications of this technology. AR app displays critical anatomical information and vitals directly onto the surgeon's field of view, helping guide incisions, highlight hidden structures, and minimize potential risks.

For surgeons, one of the biggest roadblocks is not the lack of skill, but the lack of "live" contextual information during complex procedures. With augmented reality, medical teams get a live, intuitive guide right in their field of vision instead of relying solely on themselves and the monitors.

I saw an inspiring example of this AR application from UC Davis Health, where surgeons are using augmented reality goggles to literally “see through” patients during operations. These AR glasses project 3D models of organs and tissues directly onto the patient’s body, allowing for incredibly precise navigation, almost like a GPS for surgery.

Mixed Reality Goggles Prepare Surgeons for Complex Procedures | UC Davis Health

5. Physical rehabilitation

Physical rehabilitation can be a long, frustrating process, and one of the biggest challenges for doctors is keeping patients motivated and consistent with their exercises. And AR can help with it. For patients, the technology makes rehab programs more interactive, visual, and even fun. Instead of vague instructions like “lift your arm like this,” patients can see real-time visual cues, correct their movements instantly, and feel like they’re part of a game rather than a medical routine.

This level of engagement can make all the difference in recovery speed and long-term outcomes. A meta-analysis published in the Journal of Biomedical Informatics showed that 75% of studies of AR/VR rehabilitation efficiency reported improvements among the respondents. Although AR is not yet poised to replace conventional therapies, this technology is a great complementary tool, especially for remote care interventions.

An interesting example of AR for rehabilitation is MirrorAR. It’s an AR-based app that captures patients’ motions during exercise, detects 16 key joints without any sensors, and displays the core axes of movement to help therapists analyze patients’ performance.

https://www.youtube.com/watch?v=VOaW3jXBRDE

6. Remote assistance and telemedicine

AR powers remote care assistance and telemedicine and allows experienced medical specialists to virtually “stand” next to a local healthcare provider during a procedure, offering real-time guidance, insights, or step-by-step corrections. This use of AR can change healthcare in rural or underserved areas, enabling access to high-quality expertise. After all, according to the American Medical Association, about 65% of rural areas have a shortage of primary care physicians, so getting a top specialist on-site with AR is much simpler than arranging an in-person visit.

One standout example of AR for remote assistance is Proximie, a platform that combines AR, AI, and telepresence to let surgeons collaborate from anywhere in the world. With Proximie, a remote surgeon can see exactly what the on-site surgeon sees, annotate directly into their field of vision, and guide procedures as if they were right there in the operating room. A clinical study of the tool found that the AR telementoring platform helps build medical competency safely in advanced-level surgical trainees performing complex procedures, like radical prostatectomy.

7. Patient education and self-care

I often hear from practitioners that one of the toughest parts of their jobs is explaining medical information to patients. Sure, they can nod along when doctors clarify things, but do they truly get it? AR can make a real difference in this field.

By letting patients explore 3D models displaying their conditions, AR turns passive listening into active learning. Patients can visualize muscles, bones, organs, and even how different conditions affect the body. Investing in AR for patient education has been proven to be efficient: a 12-month longitudinal study showed that patients with diabetes who were trained using AI-based AR/VR tools had 47% higher disease literacy rates.

8. Emergency response training

In critical situations, there’s no time to hesitate or second-guess; that’s why AR-based emergency response training software is such a powerful tool. It creates realistic, high-pressure environments where responders can practice diagnosing and treating patients without putting real lives at risk. This kind of immersive learning does more than traditional textbooks or lectures ever could: it builds muscle memory, sharpens critical thinking, and prepares care teams for real-world chaos.

An interesting example — although powered by VR, not AR — is Oxford Medical Simulation. Designed for training, it recreates lifelike emergency scenarios, say, cardiac arrests or multi-casualty incidents, and learners must react fast and make decisions under pressure. What I think is great about Oxford Medical Simulation is that it not only tests clinical knowledge but also communication and teamwork, which are absolutely crucial in emergencies.

Oxford Medical Simulation Demo from IMSH 2025 Exhibit Hall by HealthySimulation.c…

9. Improved diagnostics

Making an accurate diagnosis early can change the trajectory of patient care, and yet traditional imaging methods often leave room for interpretation errors. The reason for it may be that traditional CT or MRI scans compress a complex, three-dimensional reality into flat images, so it's easy to miss subtle spatial relationships or underestimate the true size of a lesion.

With healthcare AR, instead of studying static images, doctors can interact with full 3D models of organs, bones, or even vascular systems. From what I've heard from healthcare providers I worked with, this immersive perspective makes abnormalities stand out more clearly and can often catch things that might otherwise be overlooked.

10. Virtual healthcare assistants

When it comes to AR healthcare assistants, the sky is the limit. The technology can power anything from smart hospital navigation to interactive doctor onboarding assistants. And could you imagine using AR, say, for patients learning wound care at home? They could get clear, step-by-step AR guidance right when and where they need it.

To my mind, this kind of contextual, visual assistance dramatically lowers the anxiety and frustration, meaning patients and staff can focus more on doing the task, without worrying if they understood instructions correctly. And it's not just an idea, AR virtual assistants are already here. For instance, the Code Cart AR app is designed to train healthcare workers on the proper use of emergency code carts, offering hands-on learning opportunities.

Source: Reference